Chapter 8, Problem 15P

Problem

The responses of a series *RLC* circuit are

$$v_C(t) = 30 - 10e^{-20t} + 30e^{-10t}\ \text{V}$$
$$i_L(t) = 40e^{-20t} - 60e^{-10t}\ \text{mA}$$

where *vC* and *iL* are the capacitor voltage and inductor current, respectively. Determine the values of *R*, *L*, and *C*.

Step-by-step solution

Step 1 of 3

We have series RLC circuit

$$V_L(t) = 30 - 10e^{-20t} + 30\ e^{-30t}$$
$$\Leftrightarrow \quad V(t) = V_S + A_1 e^{S_1 t} + A_2 e^{S_2 t} \quad [\alpha > \omega_0]$$
$$i_L(t) = 40e^{-20t} - 60e^{-10t}\ mA$$
$$\Leftrightarrow \quad i(t) = A_1^{'} e^{S_1^{'} t} + A_2^{'} e^{S_2^{'} t} \quad [\alpha > \omega_0]$$

Comparing these equations, we get

$$V_S = 30$$
$$A_1 = -10\ ; \quad A_2 = 30\ ;$$
$$S_1 = -20\ ; \quad S_2 = -10 \quad \rightarrow (1)$$
$$A_1^{'} = 40\ ; \quad A_2^{'} = -60\ ;$$
$$S_1^{'} = -20\ ; \quad S_2^{'} = -10 \quad \rightarrow (2)$$

Step 2 of 3

Now, From equation (1) and equation(2),

$$S_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \quad And \quad S_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$
$$S_1 + S_2 = -2\alpha \quad And \quad S_1 S_2 = \omega_0^2$$
$$\left[Where \quad \alpha = \frac{R}{2L}\ ; \quad \omega_0 = \frac{1}{\sqrt{LC}} \right]$$
$$\Rightarrow \quad -30 = -2\alpha$$
$$\Rightarrow \quad \alpha = 15$$
$$\Rightarrow \quad \frac{R}{2L} = 15 \quad \rightarrow (3)$$
$$200 = \omega_0^2 \quad \Rightarrow \quad \frac{1}{LC} = 200 \quad \rightarrow (4)$$

Step 3 of 3

Also, $i(t) = C\dfrac{dV(t)}{dt} = C\left[200e^{-20t} - 300e^{-10t}\right]$

Or, $\left(A_1^{'} e^{S_1 t} + A_2^{1} e^{S_2 t}\right) \times 10^{-3}\, A = C\left\{\left[200e^{-20t} - 300e^{-10t}\right] V\right\}$

$$[S_1 = S_1^{'} \;,\; S_2 = S_2^{'}]$$

$\Rightarrow \quad 200C = A_1^{'} = 40 \times 10^{-3}$

$\Rightarrow \quad C = 200 \times 10^{-6} F \quad \Rightarrow \quad C = 200\,\mu F$

And hence, by using equations (3) and (4) we get,

$$L = \frac{1}{200C}F = \frac{1}{200 \times 200 \times 10^{-6}} = 25\,H$$

And, $R = 30L = 30 \times 25 = 750\,\Omega$